

Exhibit CC-58

A 1.1-V 10-nm class 6.4-Gb/s/pin 16-Gb DDR5 SDRAM With a Phase Rotator-ILO DLL, High-Speed SerDes, and DFE/FFE Equalization Scheme for Rx/Tx

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Abstract—A 1.1-V 6.4-Gb/s/pin 16-Gbit DDR5 is presented in 10-nm class CMOS technology. Various functions and circuits' techniques are newly adopted to improve performance and power consumption compared with DDR4 SDRAM. First, to realize two times higher speed than DDR4, the injection-locked oscillator (ILO) delay locked loop (DLL) is adopted for the low jitter high-speed performance. The proposed DLL with phase rotator (PR) and ILO allows to minimize the clock tree of DRAM, lowering skew and jitter in the DRAM internal clock path. Second, for the high-speed write operation, DQS gate opening control and write leveling are very important to minimize the turnaround time of DRAM, and thus new sequence and logic for the write-level training are introduced in this article. Third, to maximize the data valid window of read DQs, duty cycle adjustable serialize circuit methods are proposed. Finally, to improve the interface speed, the decision feedback equalization (DFE) and feedforward equalization (FFE) are adopted to Rx and Tx, respectively. By implementing all the items mentioned earlier, the 1.1-V 6.4-Gb/s/pin 16-Gbit DDR5 achieved 6.4-Gb/s/pin performance at 1.05-V V_{DD} , with its power bandwidth efficiency 30% higher than that of DDR4.

Index Terms—DDR5 decision feedback equalization (DFE), DDR5, DIMM, DRAM interface, DRAM, feedforward equalization (FFE), on-die termination (ODT), phase rotator (PR) delay locked loop (DLL), write-level training.

I. INTRODUCTION

THE widespread use of the Internet, mobile devices, and Internet-of-Things (IoT) technologies as well as the processing of big data [1] gave rise to the high-performance server system, cloud computing, and edge computing. The performance requirements of these server and computing systems require DRAM to improve its bandwidth, capacity, and power. In response to this demand, DDR5 has been proposed as the next-generation device of double data rate (DDR)

TABLE I
DIFFERENCE BETWEEN DDR4 AND DDR5

	Bank Group (BG)	Banks in BG	On chip ECC	Voltage Regulator Position	Prefetch	Max. Data rate	VDD	VPP
DDR4	4	4	-	Outside of DIMM	8	3.2Gb/s/pin	1.2V	2.5V
DDR5	8	4	128:8 SEC	VR on DIMM	16	6.4Gb/s/pin	1.1V	1.8V

SDRAM with various new functions to overcome scale and speed limitations. The difference between DDR4 and DDR5 is summarized in Table I. DDR5 adopted the on-die error correcting code (ECC), such as LPDDR4, to overcome the scale limit [2]. The on-die ECC allows to support the data retention time of the DRAM and to improve the defects caused by the TID [3]. Some asynchronous parameters have been relaxed to improve the reliability of the DRAM. Its write recovery time (tWR) is two times longer than that of DDR4, and it also increased tRCD/tAA/tRP as well. The methods applied to overcome DRAM scale limits may increase latency and degrade system performance. However, the number of banks and bank groups of DDR5 has doubled [4] compared with that of DDR4. The same bank refresh function is adopted to access a bank, while other banks in the same bank group are refreshing. These features compensate for the increased latency and allow to maintain the system bandwidth.

To increase the system throughput, DIMM configuration is changed to two channels, as shown in Fig. 1(a). DDR4 has 24 command and address (C/A) pins, 64 DQ pins, and 16 DQS pairs in 1 DIMM. DDR5 separates 1DIMM into two channels. CB0–CB7 are DRAMs for ECC, as shown in Fig. 1(a). The two-cycle command is applied to reduce the number of C/A pins per channel from 24 to 14. While CKE and on-die termination (ODT) pins are removed, the ODT control method is changed from the pin-based control to the non-target read/write ODT command-based control. While CKE pins were previously used to control the power-down

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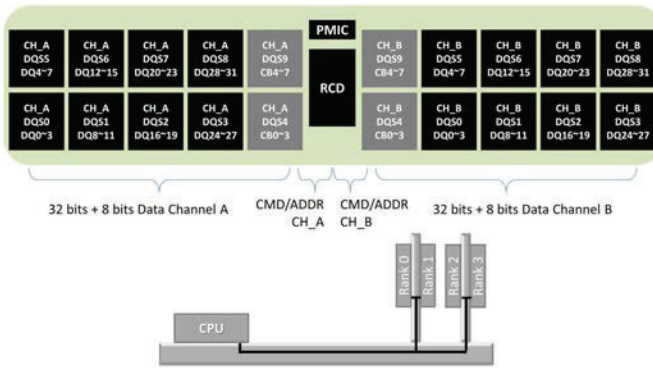


Fig. 1. DIMM configuration of DDR5. (a) Top view of DDR5 DIMM. (b) Side view of DDR5 configuration.

mode and self-refresh mode entry and exit, now, these modes are controlled by commands. In case of DQ pins, the prefetching size is doubled to compensate for the reduced number of DQ pins in the channel. The power management integrated circuit (PMIC) is mounted inside the DIMM for a stable operation.

DDR5 doubled the target speed while lowered the supplied voltage compared with DDR4. DDR4 operates at 3.2 Gb/s/pin at 1.2 V, while DDR5 operates at 6.4 Gb/s/pin at the supplied voltage of 1.1 V. In case of DDR4, two DIMMs are connected to one channel in general, and one DIMM is composed with one or two ranks. In this case, four DRAMs can be connected to one multi-drop channel, as shown in Fig. 1(b) [5]. Unlike LPDDR4 that generally uses package on package (POP), reflection and crosstalk become a bottleneck in the high-speed operation of DDR5 with DIMM topology as multiple DRAMs are connected to the multi-drop channel. To overcome the speed limit of the multi-drop channel, DDR5 adopted the four-tap decision feedback equalization (DFE) with a new write training method to improve the receiver margin, which is described in Section III.

This article is organized as follows. Section II describes the overview of the DDR5 architecture. Section II explains the four-tap DFE and write training method that are adopted to improve the Rx margin. The injection-locked oscillator (ILO) delay locked loop (DLL), duty cycle adjuster 4:1 serializer, and 1-tap feedforward equalization (FFE) for Tx margin are explained in Section III. The measurement results of this article are shown in Section V. Section VI summarizes this article.

II. DDR5 ARCHITECTURE

As shown in Fig. 2, DDR5 introduced several new schemes in its architecture to overcome limitations of DDR4. Its burst length is set at 16 for bandwidth enhancement. As the overall top chip architecture is shown in Fig. 3, the number of bank groups is extended from 4 to 8, with the number of banks reaching 32, which is twice that of DDR4. This architecture compensates for the asynchronous parameter delay caused by the technology scaling limit and increases the system throughput at the same DRAM bandwidth. Without enhancing the pre-fetch size and the number of banks, it is hard to achieve a higher bandwidth, even if the bank group interleave function is adopted.

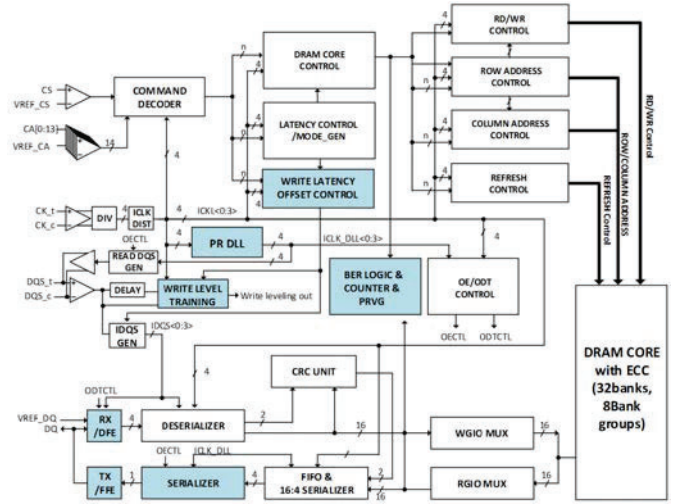


Fig. 2. DDR5 functional block diagram.

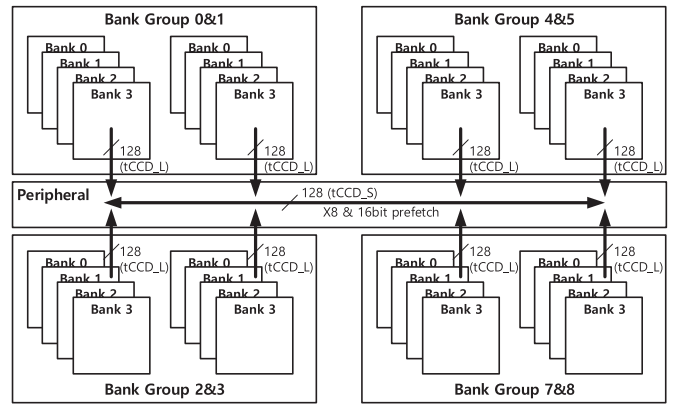


Fig. 3. DDR5 top chip architecture.

In addition, a new Rx/Tx equalization scheme is applied in order to maintain the signal integrity of interface at the higher bandwidth of the multi-drop channel. We also need to overcome the channel noise, such as reflection and the inter-symbol interference (ISI), to reach beyond the data rate of DDR4, 3200 Mb/s, and that is why the DFE and FFE are implemented in Rx and Tx, respectively, as shown in the bottom left of Fig. 2.

Synchronization between DQS and clock may cause a serious problem in the high-bandwidth memory subsystem. Due to the slow migration of the DRAM process scaling, the internal delay difference between the clock and the DQS signal would create a speed bottleneck in modern DRAM design. For high-speed operation, it is necessary for DRAM receiver circuits to reduce the internal delay, which causes the timing jitter. Therefore, a new write leveling training scheme is implemented to get clean internal DQS of DRAM using the DQS gate open timing trained by the system. It allows a stable operation of DQS with a frequency divider to enhance the internal margin of write data.

In addition, it is important to reduce the jitter of on-chip clock tree, including the DLL to improve the read tDV, and that is why a new type of DLL is implemented. It is comprised of a phase rotator (PR) and ILO. It allows a shorter clock tree

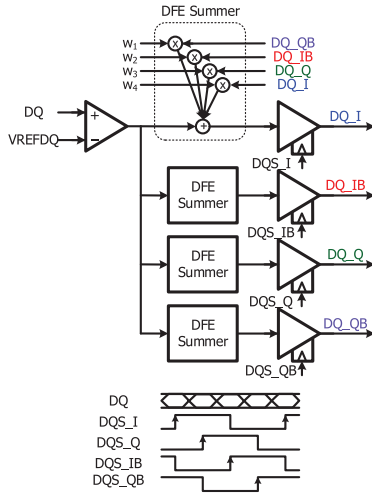


Fig. 4. Block diagram and timing diagram of DFE.

than the conventional DLL and the self-duty correction by oscillation, which will reduce skew and jitter of the output data.

III. WRITE OPERATION

A. DFE

To support a high-density memory system, the main memory supports multiple memory modules and has multi-drop topology for interconnect. To increase system bandwidth, the data rate of the memory is also important. In the multi-drop topology, the ISI noise by reflection is a bottleneck to increase the interface speed. The reflection noise can be effectively removed with the DFE [6], [7]. However, DDR4 SDRAM does not support the DFE, and it is difficult to achieve the data rate of above 3.2 Gb/s. In the DDR5 SDRAM, the four-tap DFE is defined to overcome the data rate limitation of the multi-drop channel topology. To minimize the overhead of the DFE, the DFE adaptation is performed at the memory controller by setting the mode registers for DFE tap coefficients. Fig. 4 shows the block diagram of the DFE. It has a four-way interleaving architecture by using quad-rate four-phase internal DQS signals (DQS_I, DQS_Q, DQS_IB, and DQS_QB). As the four-tap feedback data of each slicer are different, each slicer has a separate DFE summer.

The DFE summer adopts CML-type circuits, as shown in Fig. 5, to support various data rates of the DDR5 SDRAM. The current-integrating-type summer is widely used to reduce the DFE feedback delay and to increase the signal-to-noise ratio (SNR) [6], [7]. However, it requires an additional calibration sequence to normalize integrating gain for various data rates. The DFE has four taps, and each tap coefficient (w_1 , w_2 , w_3 , and w_4) is set by the mode registers. The range of each tap coefficient is defined by signal integrity analysis of the DDR5 DIMM configuration, and the first post-cursor is the most significant value. w_1 , w_2 , w_3 , and w_4 have 7, 5, 5, and 5 bit, respectively; 1 bit is for sign bit and the other bits are for the magnitude of the tap coefficient. Each DFE tap coefficient is converted to current by the current digital-to-analog converter (DAC) according to the tap coefficient.

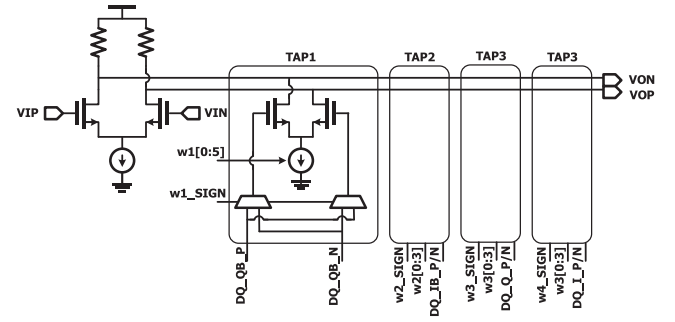


Fig. 5. Circuit diagram of DFE summer.

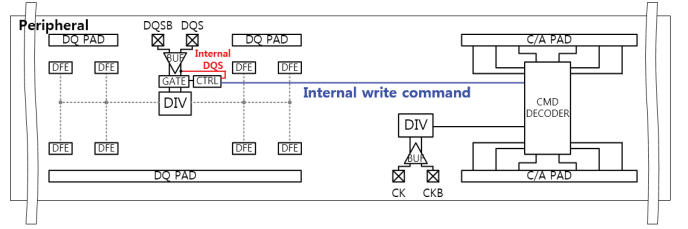


Fig. 6. Block diagram of peripheral for write operation.

The magnitude is controlled by the digitally controlled current source, and the sign is controlled by switches with polarity of the feedback data. The converted current is applied to the main signal through the load resistor of the summer.

B. Write Training

DDR5 has changes in the write-level training for a stable DFE operation. The write-level training is divided into two sequential processes. The first process is the write leveling to match the CLK and DQS (Data Strobe) signals, and the second process is the tDQS2DQ training to match the DQS and DQ as they are unmatched inside the DRAM unlike the DDR4. Fig. 6 shows the block diagram of peripheral for write operation.

The delay difference between the DQS path and the DQ path in Fig. 6 is synchronized through the tDQS2DQ training. The DQS path consists of the DQS buffer, gating circuit, divider, and DQS tree with repeaters, and the DQ path includes the variable-gain amplifier (VGA) and summer. As DDR4 matches the DQ and DQS inside the DRAM, the matching delay is added to the DQ path [8]. However, DDR5 removes the matching delay in the DQ path to minimize the feedback delay of DFE. Thus, DDR5 is required to perform the tDQS2DQ training per pin due to the mismatch of the DQS and DQ. Its disadvantage is the Rx setup/hold jitter caused by tDQS2DQ delay changes depending on the voltage and temperature variation. Therefore, when using the unmatched DQS and DQ relationship, it is important to reduce the tDQS2DQ delay.

For a system with the DQS, it is important to gate the DQS by the burst length after the DRAM receives a write command [9]. In order to gate the DQS, the DQS needs to be matched with the internal write command that is synchronized to the internal clock, but the write command-decoding delay is more than several clock cycles at high speeds. The delay difference between the DQS and the internal command is

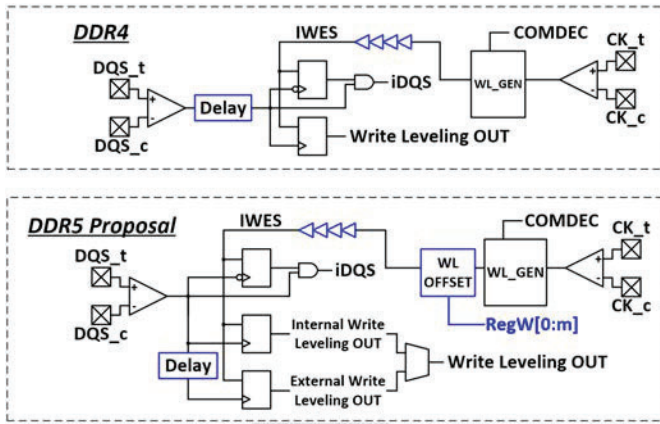


Fig. 7. Write-level training: block diagram about write leveling scheme on DDR4 and write-level training scheme on DDR5.

shown in Fig. 6. As for DDR4, write leveling compensates for the CLK to DQS timing skew, but it only compensates the timing skew generated from the outside of the DRAM. Therefore, to gate the DQS using the internal command in DDR4, a delay is inserted internally in the DQS path to match with the internal command, as shown in Fig. 7. However, as for DDR5, it is not feasible to add a delay to the DQS path because an additional delay will cause the jitter that interferes the high-speed operation, as mentioned in the tDQS2DQ training section. To solve this clock-domain-crossing problem, caused by the delay difference between the CLK and DQS, a new write-level training method for the unmatched CLK-to-DQS path is proposed. Fig. 7 shows the block diagram of a DDR5 write with a DQS control scheme. While the conventional write leveling synchronizes the CLK and DQS at the external pin of the DRAM, the write-level training of DDR5 synchronizes the CLK and DQS at the DQS gating point inside the DRAM.

Fig. 8 shows a flowchart of the new write-level training sequence. The first step is the external training such as DDR4 using the matching delay in the DQS path described in Fig. 7. This step sets the reference point of the multirank DRAM. The delay difference in the DQS and CLK path is different for each DRAM connected with one rank because each DRAM has different characteristics depending on the process skew or vendor, and it causes a bubble in the channel in the rank interleave operation.

The second and third steps are for the internal training. The internal training steps do not use the matching delay in the DQS path, as shown in Fig. 7. The second step is the coarse training, pulling the internal write enable signal (IWES) by tCK using the mode register write (MRW) setting. The result of the coarse training is stored in the “WL internal cycle alignment” mode register. When the controller sets the WL internal cycle alignment bits using the MRW, the DRAM reduces the shifting amount of the shift register that generates the IWES according to the MRW value. The final step is fine training, sweeping the DQS signal to find the rising edge of IWES. After the training, the IWES and DQS are matched without any additional delay. Therefore, it saves the power

consumption of the DQ/DQS delay and improves setup and hold characteristics.

Fig. 9 shows the relationship between the IWES and the internal DQS depending on the write preamble. To maximize the domain crossing margin, the IWES needs to be located at the center between the beginning of the preamble and the first falling edge of the first preamble. As the write-level training is synchronized with the IWES and DQS, the DQS should be shifted by $1.25t_{CK}$ ($t_{WL_ADJ_END}$) as shown in Fig. 9 in the $2t_{CK}$ preamble mode. After the fine training step (third step) in write leveling training, the controller delays the DQS by $1.25t_{CK}$. To compensate the $t_{WL_ADJ_END}$ delay, the DQS should be shifted by $-0.75t_{CK}$ ($t_{WL_ADJ_START}$) before the internal training step (second step). Since the $t_{WL_ADJ_START}$ and $t_{WL_ADJ_END}$ are applied, the final DQS position is within $\pm 0.5t_{CK}$ of the reference point (result of external training) of the write-level training.

C. ODT Control

The ODT has been adopted for a better signal integrity since DDR3. DDR4 has two ways to control the ODT. One is to change to RTT_{NOM} using the ODT pin, and the other is the dynamic ODT method to change to RTT_{WR} using the write command. The dynamic ODT, controlled by the DRAM, changes the RTT value from RTT_{PARK} to RTT_{WR} during the DQS and DQ toggle. The ODT pin, controlled by the DRAM controller, changes the RTT value from RTT_{PARK} to RTT_{NOM} . As terminating the non-target DRAM improves SI of the target DRAM in the multirank system [10], DDR4 uses the dynamic ODT for the target rank during the write operation and utilizes the ODT pin controlled by the controller set the non-target rank to RTT_{NOM} . On the other hand, as DDR5 removed the ODT pin as previously mentioned, a new command is created to control the non-target ODT. Fig. 10 shows the timing diagram for the DDR5 ODT control. DDR5 adopts the 2-cycle command, and the write command applies the write command to C/A pin and sets CS pin at low in the first cycle and at high in the second cycle. The non-target write ODT command sets the CS pin at low for both the first and second cycles with the write command.

Target and non-target ODT commands change the RTT value, while DQ conducts toggling. After the write command is issued, the RTT value changes after the CAS write latency (CWL) clock and until the end of burst length. As the CLK, DQS, and DQ are all mismatched in DDR5, it is necessary to move up or push back the “target/non-target write RTT enable time” and “target/non-target write RTT disable time” to cover the applied data with RTT_{WR}/RTT_{NOM} . The ODT timing should be adjusted based on the relationship among CLK, DQS, and DQ. Fig. 10 shows the ODT timing of DDR5 and the block diagram for ODT control. The mode register latch stores $RTT_{WR}/NOM/PARK$ values and stores the ODT timing offset information. When the write or NTODT command is issued, pulse is generated to cover the data based on the ODT offset information. If the command is the write command, the signal output will be WTT_{WR} value. If it is the NTODT command, the output will be WTT_{NOM} value. If there is no command, the output will be RTT_{PARK} value.

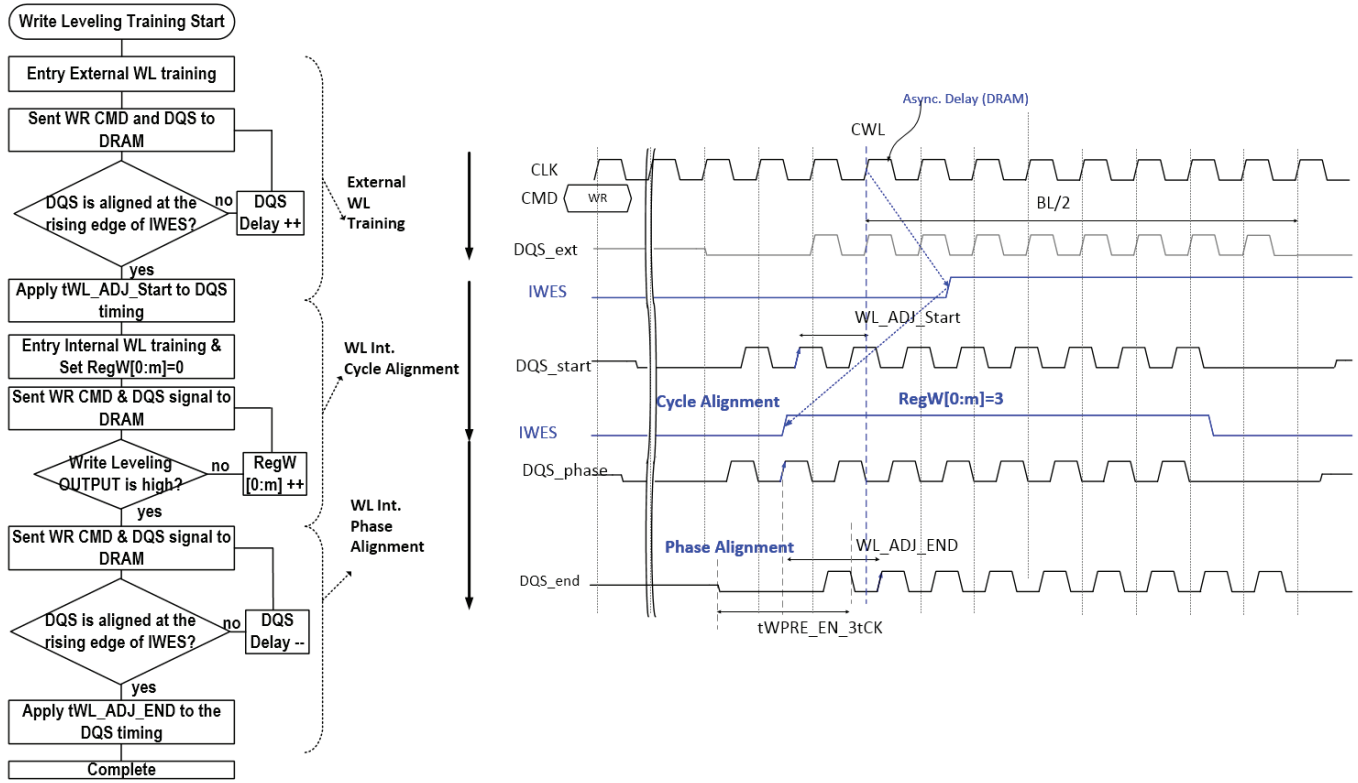


Fig. 8. Flowchart and timing diagram of write-level training.

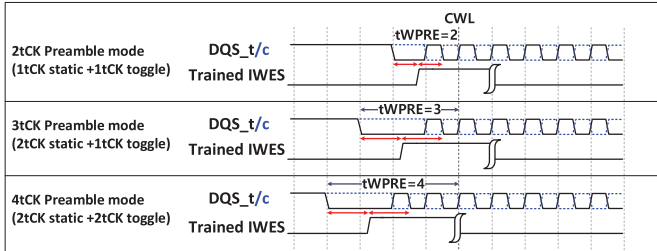


Fig. 9. Relationship between the IWES and the internal DQS.

Since DDR5 has an unmatched relationship between the DQS and DQ, the DQS toggle does not match the DQ toggle, as shown in Fig. 10. Thus, DDR5 separated the ODT control of DQS and DQ. The RTT value of DQS is DQS_RTT_PARK and is held static during the entire write operation.

IV. READ OPERATION

A. ILO DLL

A conventional DRAM DLL controls the clock delay by changing the number of delay elements to synchronize the external CK and DQS. A NAND-based delay element is typically used to delay the clock signal, and the delay is up to a few nanoseconds [11]. As the clock speed increases, a large number of delay elements increase the clock jitter and duty cycle distortion. In addition, when a burst read command is issued, the VDD suddenly drops, and the skew of the CK and DQS becomes large due to the large delay variation of delay line until the DLL tracks the skew.

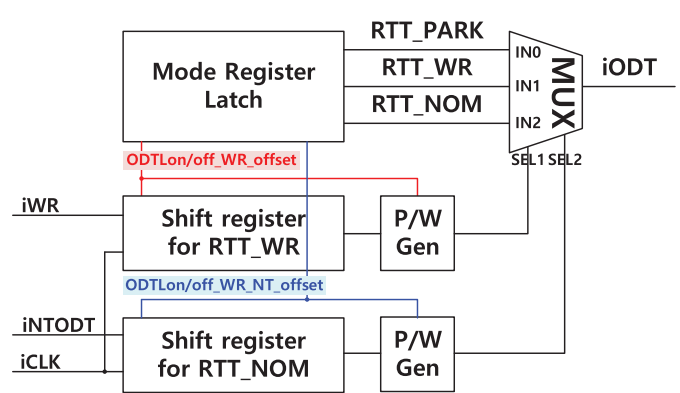
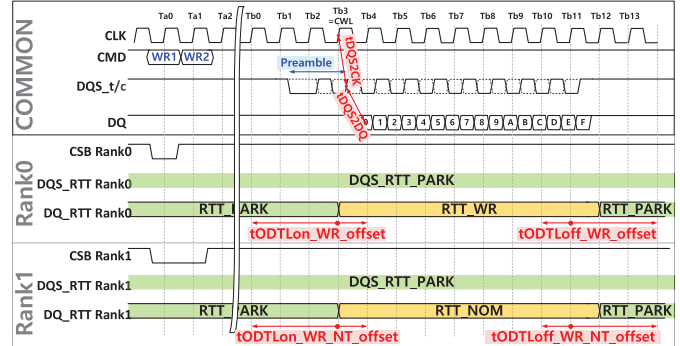


Fig. 10. Timing and block diagram for DDR5 ODT control.

To address this problem, the proposed DLL adopts a PR and an ILO, as shown in Fig. 11 [12], [13]. The external CK is sent into the clock buffer and divider, and then, quadrature

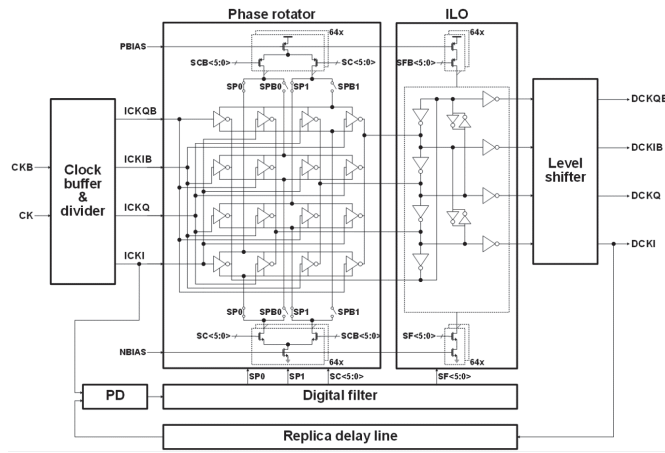


Fig. 11. Implemented DLL based on a PR and an ILO.

clocks are generated. The PR interpolates between the input quadrature clocks and injects the output quadrature clocks into the ILO. The PR achieves 1-/64-UI steps using the 6-bit current DAC. To reduce switching glitches, the PR controls the amount of current to each inverter, instead of switching the input clock phases. It achieves the infinite phase rotation without delay elements and reduces the delay variation caused by the supply noise.

The ILO consists of a current-controlled two-stage differential ring oscillator, with the input quadrature clocks injected into its four oscillation nodes. A conventional PR with the RC loading has a large RC time constant, decreasing the noise immunity. The proposed architecture adopts the active resonant load instead of the RC load. The ILO behaves like an active bandpass filter and provides a good phase linearity with jitter improvement [14]. The ILO also improves the PR linearity and reduces the quadrature-clock-phase error like an active polyphase filter [15].

A replica-delay cell-based frequency tracking loop is utilized to compensate the frequency error between the injection clock and the ILO's free-running clock [16]. By comparing the phase difference between the replica-delay cell output and ILO's output, the ILO's free-running frequency is adjusted by the 6-bit current DAC during the initial training period. The lock range of ILO is designed to be wide enough to cover the voltage and temperature variation after initial frequency training. When the frequency of input clock changes, the free-running frequency of ILO needs to be retrained. The remaining quadrature clock skew is compensated by the duty cycle adjuster (DCA) placed after the ILO during the training period.

B. FFE

The FFE is widely used in interface circuits to improve operating speeds. In the DRAM process, however, the conventional FFE schemes are infeasible as it is difficult to implement a full-rate flip-flop. Therefore, DRAM has used the FFE by adding an additional segment to emphasize the fixed gate delay for the beginning period of data transition. This is simple to implement but has many drawbacks. The amount of emphasizing delay is fixed regardless of the data

rate, and high level emphasizing is not feasible in the pseudo-open drain (POD) signaling. Moreover, adding a segment for emphasis not only increases the total output capacitance but also causes the impedance mismatch. Fig. 12 shows the proposed FFE circuit optimized for the DRAM process and its waveform. Fig. 12(a) shows the FFE concept for the DRAM process. A combination of the MUX and serializer allows to realize the FFE without the full rate flip-flop. The overall architecture is shown in Fig. 12(b). The MUX in each segment is followed by two common 4:1 serializers to reduce the total number of serializer, which consumes much power. One serializer is for the current data, and the other is for the data of the previous cycle. Basically, since impedance should be matched with a fixed value when identical data are consecutively transmitted, the target impedance is divided and sent to multiple parallel segments, each of which is controlled by the MUX to change the tap coefficients of the FFE coarsely. Above 1-UI setup and hold margin can be obtained by the quarter-rate FFE. For a finer FFE control, we add fine FFE codes in the main seven segments at a cost of a small impedance mismatch.

C. READ Serializer

DDR5 has a high-speed serializer for multiplexing four data inputs. In order to operate at a higher speed (max 6.4 Gb/s) than DDR4 (max 3.2 Gb/s), the DDR5 serializer uses four-phase divided clock signals, CLK0, CLK1, CLK2, and CLK3, as shown in Fig. 13. Each data signal, D0, D1, D2, and D3 in Fig. 13, is latched and serialized at the rising edge point of each clock signal.

The serializer has four-data latches, and each data latch comprises a cross-coupled latch and a tristate-type driver. The four-phase CLK controller block in Fig. 13 is configured, as shown in Fig. 14. The four-phase CLK controller in Fig. 14 cuts the input clock into a next phase clock to produce a 1-UI pulse signal that connects to each data latch in Fig. 14. When the input clock signals are at the "H" state, the latch drives UP/DN with the input data signal. Otherwise, the latch is pre-charged and the floating UP/DN allows the next-stage latch to operate.

To get a stabilized operation at higher speed, the clock signal controller has a clock duty controller that changes the clock duty cycle using specific codes. By enabling the additional transistor in the controller according to the level of the clock signal and the specific codes such as Duty_H0 and Duty_L0 in Fig. 14, the rising and falling slope of the clock signal can be changed by the codes, and the clock controller can adjust the clock signal duty, which can also correct the DQ output skew.

This specific code value can be loaded externally or a pre-test can be used to find a code value that can correct the DQ skew. If this value is stored in memory via fuse, it will be recalled in the next operation so that the DQ skew can be corrected.

D. On-Die Write and Read I/O BERT

The test methodology for high-frequency random deterministic jitter, duty cycle error, and bounded uncorrected error is

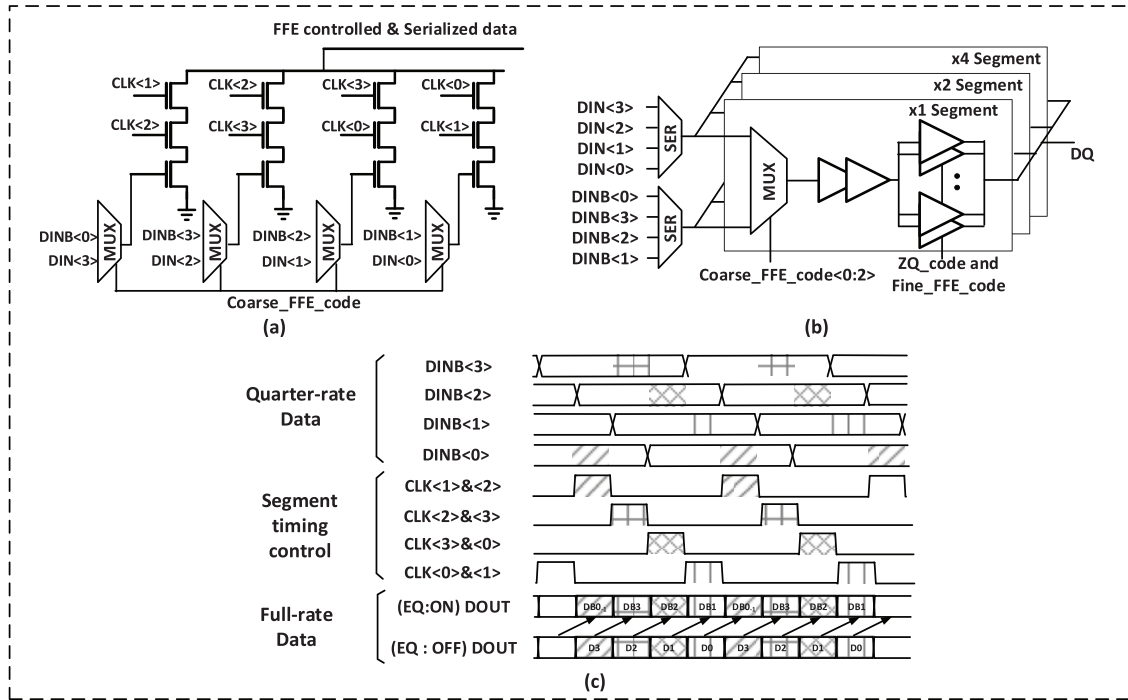


Fig. 12. Proposed FFE block diagram and waveform. (a) DRAM optimized FFE concept. (b) Overall transmitter architecture. (c) Timing diagram of DRAM optimized FFE transmitter.

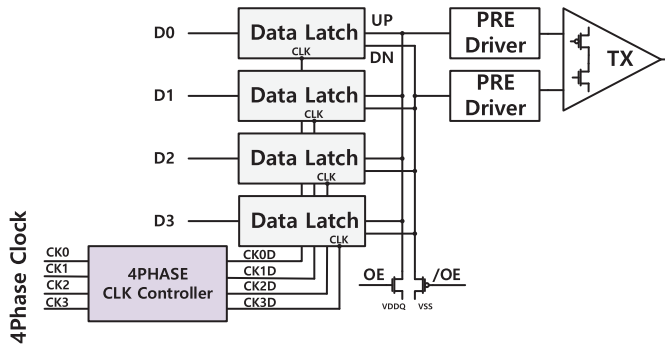


Fig. 13. 4:1 data multiplexer.

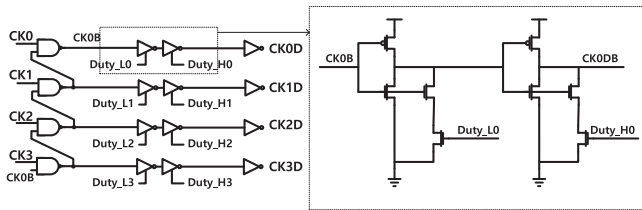


Fig. 14. Four-phase clock controller block.

required for the single-ended signaling of the DDR5 system, and that is why the on-chip BERT scheme is applied to measure the performance of the DRAM I/O.

Fig. 15 shows the architecture of the on-chip bit error rate (BER) circuit for the DDR5 system. In the proposed DDR5 system, there are two modes: the BER Tx mode and the BER Rx mode. The role of the BER Tx mode in Fig. 15 is to measure the performance of transmitter and read

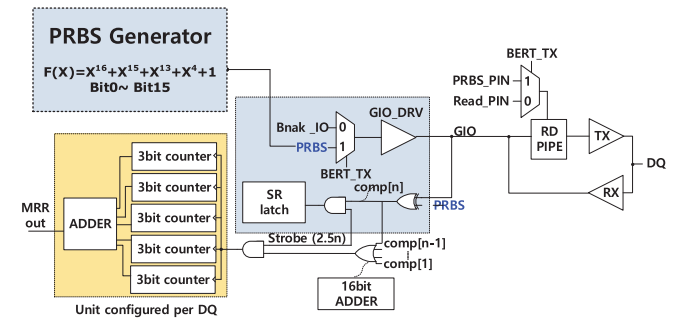


Fig. 15. On-die BERT block diagram.

of the DRAM. It makes and sends agreed pseudo-random patterns to the host. If the host issues the MRR command on the BER Tx mode, the DRAM transfers premised data as the mode is set to pre-determined pattern or PRBS. The purpose of the BER Rx in Fig. 15 is to measure the performance of receiver and write of the DRAM. The data from the DQ pads are compared with the PRBS data of the DRAM. The BER system detects data errors if they occur in the channel, receiver, or write of the DRAM. The detected errors that retimed by the strobe go into the counters. The five 3-bit counters store and send the number of errors to the adder. The output of the adder indicates all error occurrence numbers, and it can be read by the MRR command. After the host issues the MRR command and waits for the CAS latency, the final calculated value is loaded into the global IO to transfer data to DQs. Thus, the values can be read out from the BL0 of the LSB up to BL15 of the MSB based on the error counting limitation setting. The on-chip BER circuit located at each DRAM chip can compare all DQ data and count up to $2^9 - 1$ bit error.

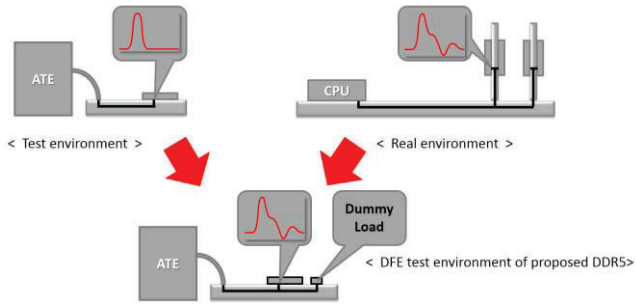


Fig. 16. Test environment of proposed DDR5.

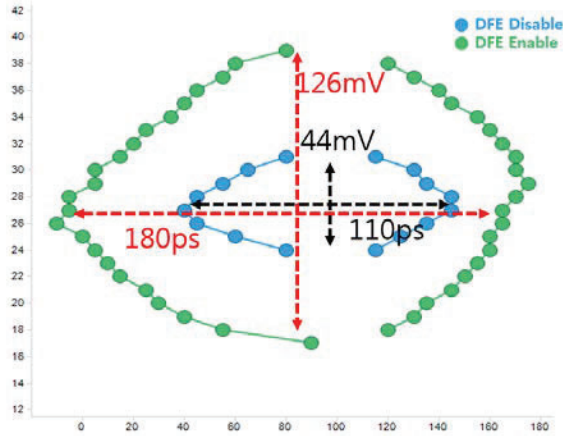


Fig. 17. Measurement result of four-tap DFE.



Fig. 18. BERT Result of Rx.

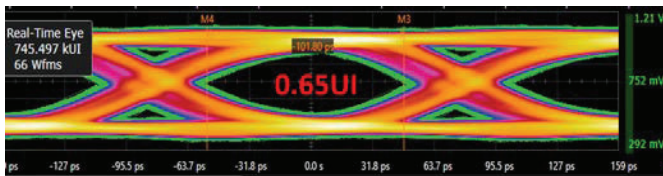


Fig. 19. Stressed eye of Tx.

V. MEASUREMENT RESULT

As the test environment of the proposed DDR5 shown in Fig. 16, the DFE is applied to cancel the reflection noise of the channel. However, the reflection noise does not occur in a general test environment. To evaluate the four-tap DFE in an environment with the reflection noise, a new test environment was developed, as shown in the bottom of Fig. 16. The dummy load generates the reflection noise similar to the real application environment. As shown in Fig. 17, by using



Fig. 20. tCK versus VDD shmoo.

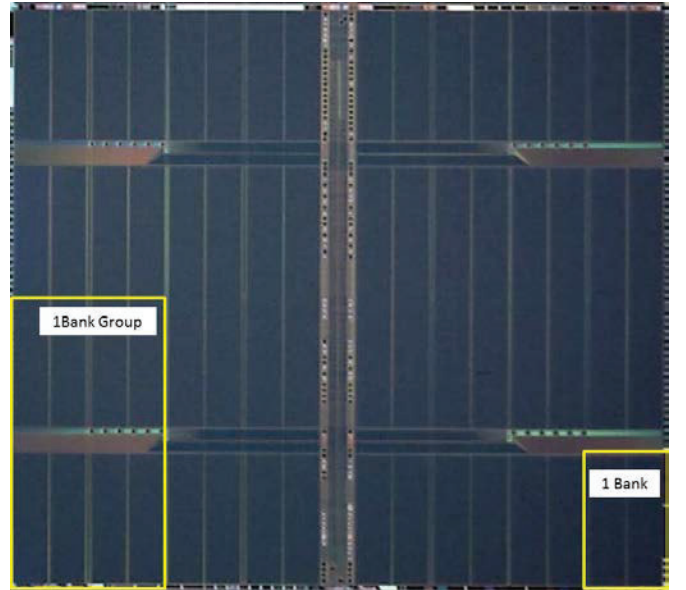


Fig. 21. Die micrograph.

TABLE II
CHIP SUMMARY

Chip summary	
Technology	1y-nm 4-metal DRAM process
Data Rate	6.4 Gbps/pin
IO configuration	X4/x8/x16
Burst length	BC8, BL16 on-the-fly
Chip area	76.22mm ²
Supply voltage	VDD/VDDQ 1.1V, VPP 1.8V
RAS feature	In DRAM ECC
Equalize scheme	DFE/FFE

four-tap DFE, the eye height was improved from 44 to 126 mV, and the eye width was improved from 0.48 to 0.79 UI. This result was measured at 1.1 V, 4.4 Gb/s, and random

pattern with the DFE training. The transceiver measurement result of this article measured under 6.4 Gb/s and 1.067 V is shown in Figs. 18 and 19. Rx was measured at the PRBS-7 pattern, and the BER9 and 1DQ measurement results show 0.64-UI window, as shown in Fig. 18. Tx was measured at the PRBS-8 pattern and the 1DQ measurement result shows a 0.65-UI window, as shown in Fig. 19. Fig. 20 shows the frequency versus voltage shmoo graph. This article achieves 6.45-Gb/s/pin data rate with 1.05-V supply voltage. The 16-Gb DDR5 implemented the 10-nm class four-metal layer DRAM CMOS. The overall die microphotograph is shown in Fig. 21. A summary of this article is shown in Table II.

VI. CONCLUSION

In this article, a 6.4-Gb/s DDR5 SDRAM is realized with a 10-nm class CMOS four-metal process. DDR5 SDRAM adopts lower supply voltage to reduce power consumption compared with the previous DDR4 SDRAM. As a result, the power efficiency is improved by about 35% compared with DDR4. There are various features to guarantee stable transaction at a higher speed, such as 6.4 Gb/s. For the extended system throughput, 32 banks, 8 bank groups, and burst length 16 are adopted, and for the high-speed read operation, the ILO DLL and dynamic 4:1 read mux are implemented. A new write-level training, DFE/FFE equalization scheme for Rx/Tx, BER self-measure circuits are also developed for the BER test, and the four-tap DFE improves the Rx eye. As a result, all these features contributed to achieving the measurement result of 6.4 Gb/s at 1.05 V.

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